

219/504MAN RESOURCE MANAGEMENT

Managing Risk and Recovery

Week 6 – Session 3

Register



Intended Module Learning Outcomes (LO's)

By the end of the module students should be able to:

LO1: Appraise the development of the human resource function.

LO2: Critically examine the traditional and contemporary approaches of recruitment and selection.

LO3: Analyse the procedures and practices of performance management and remedial activities.

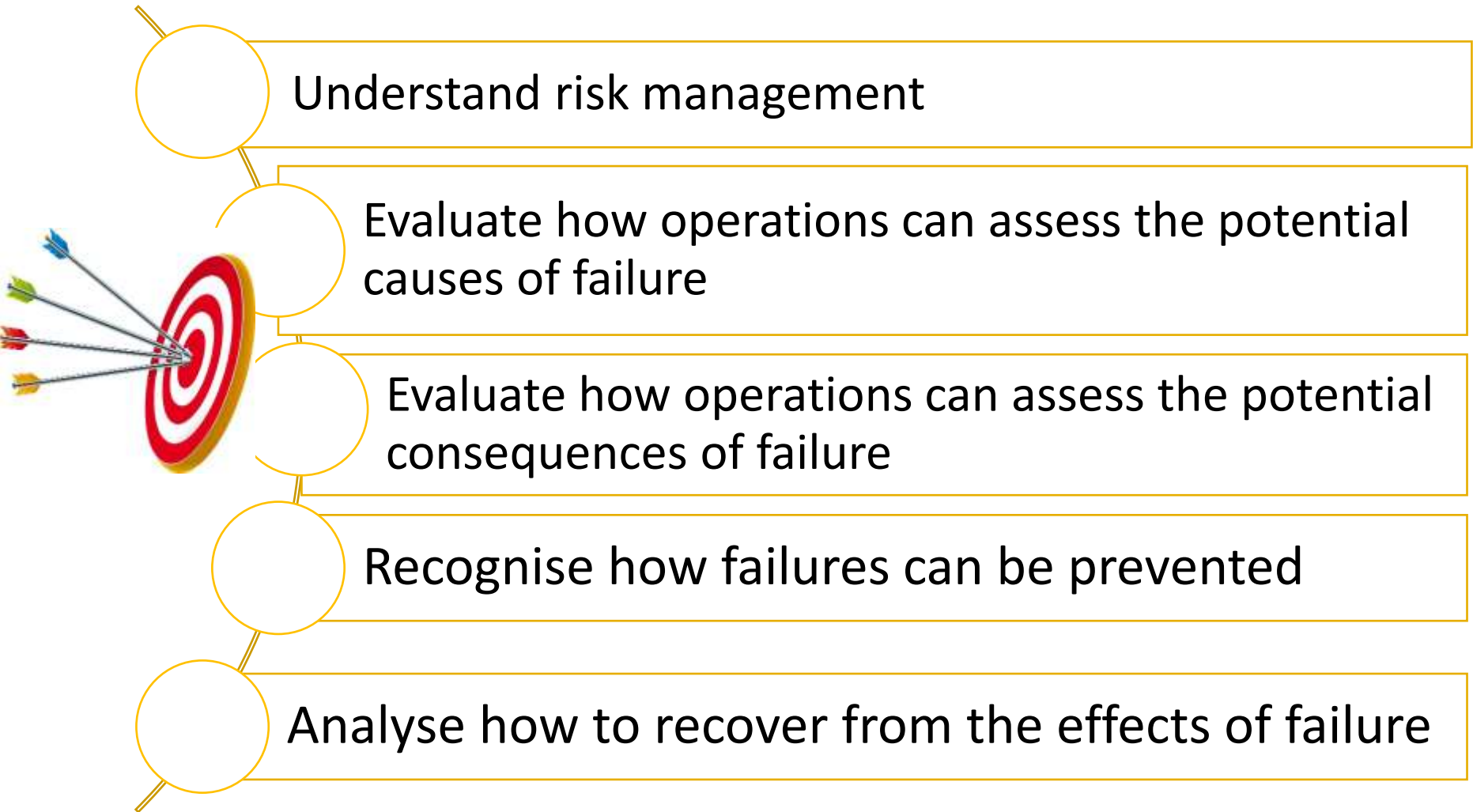
LO4: Investigate and plan the resources required to achieve an organisation's aims and objectives.

LO5: Evaluate and appraise the use of resources against organisational plans.

LO6: Formulate plans to ensure that resources are maximised and remedial actions taken where appropriate.

Session Outcomes

At the end of this session learners should be able to:



Warm – up Think, Pair & Share



Activity

5 minutes

What is risk management?



Watch & Think-Pair-Share



Activity

10 minutes

What exactly is risk management?

<https://www.youtube.com/watch?v=BLAEuVSAIVM>

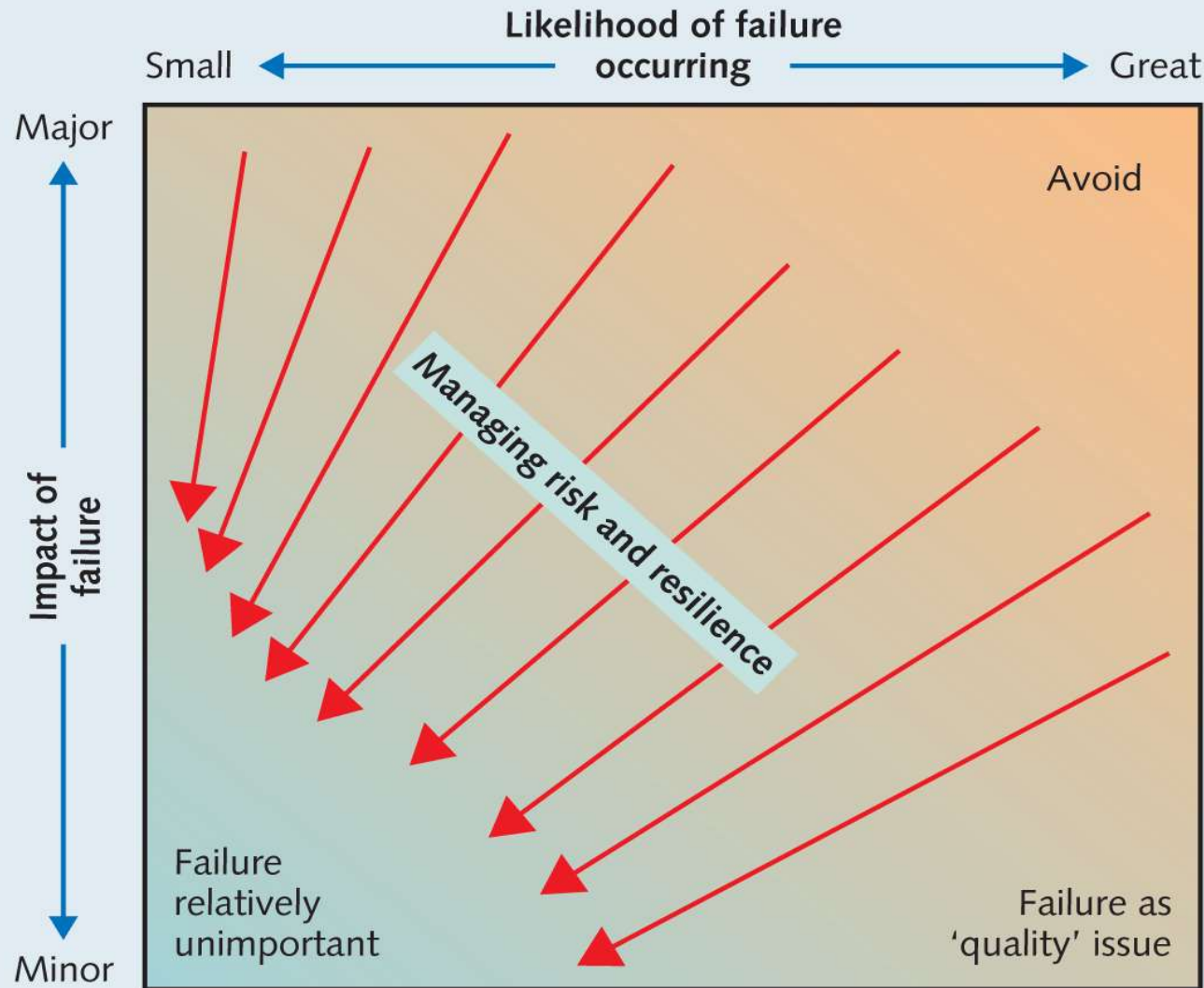


Risk management basics: What exactly is it?

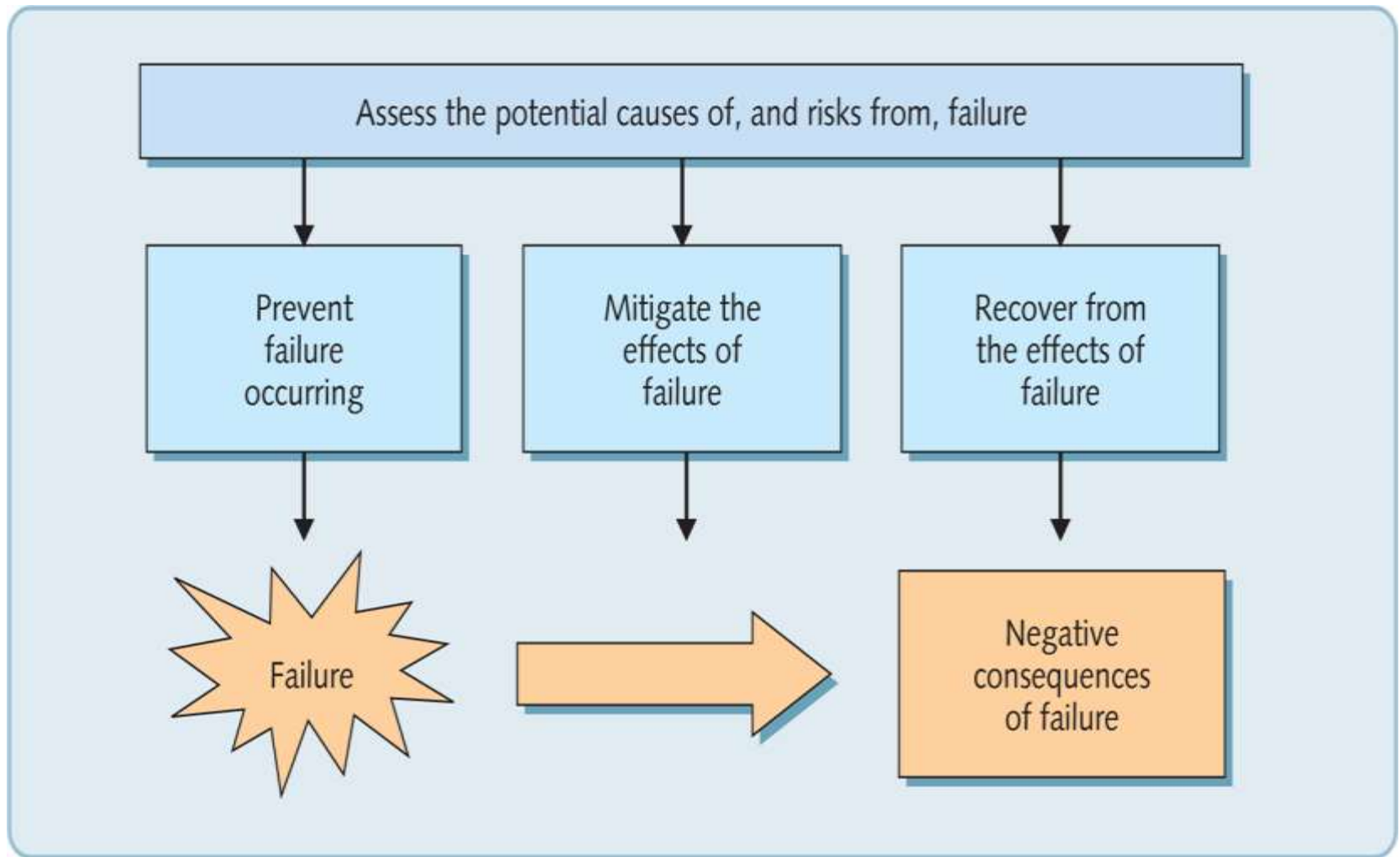
Failure will always occur in operations; recognising this does not imply accepting or ignoring it

Resilience is governed by the effectiveness of failure prevention, mitigation and recovery

How failure is managed depends on its likelihood of occurrence and the negative consequences of failure

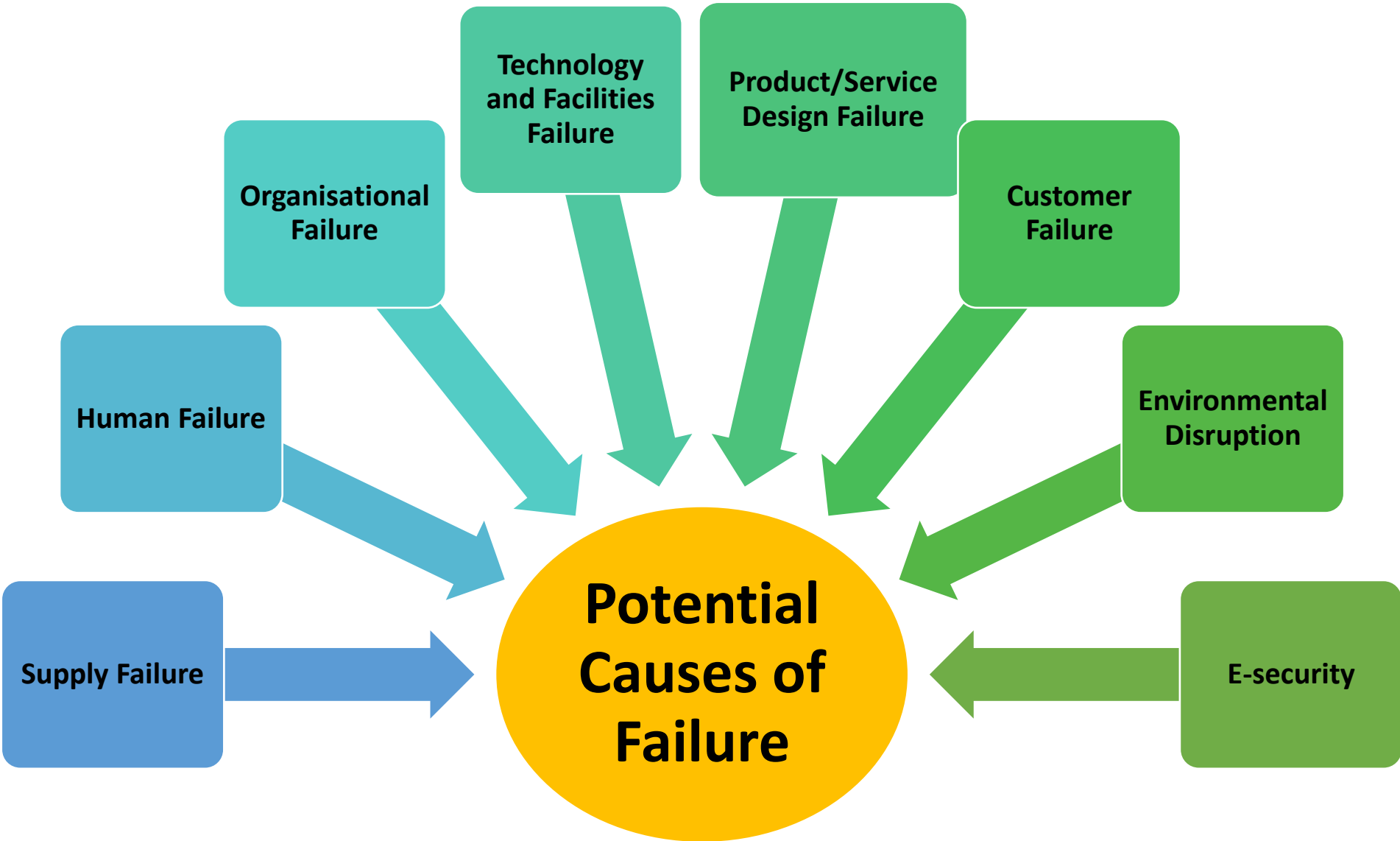


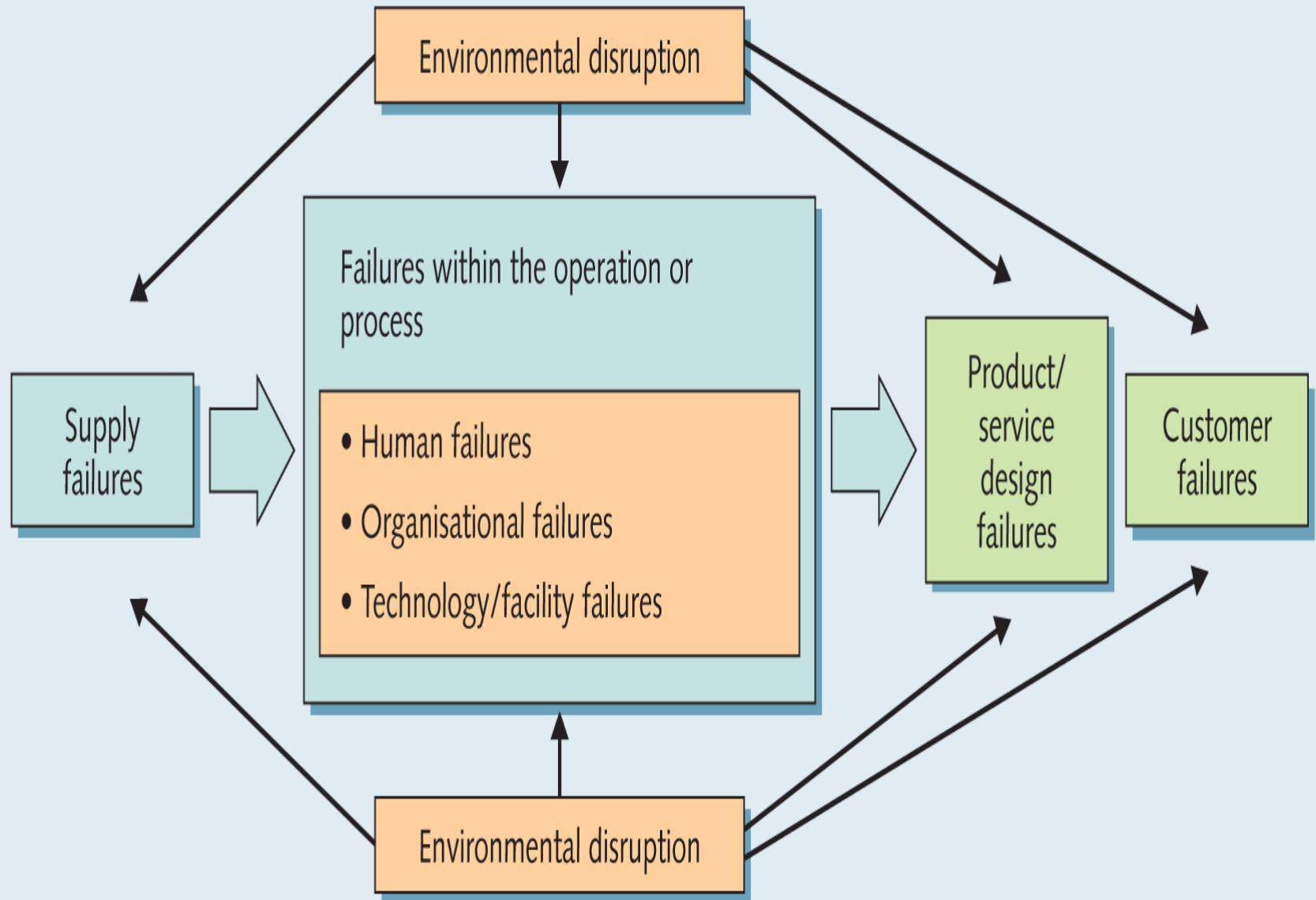
(Slack and Brandon-Jones 2018)



Risk management involves failure prevention, mitigation the negative consequences of failure and failure recovery

A 'failure to
understand
failure' can be
the root cause
of a lack of
resilience





Post-failure
Analysis

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graph LR; A[Post-failure Analysis] --- B[Accident Investigation]; A --- C[Failure Traceability]; A --- D[Complaint Analysis]; A --- E[Fault-tree Analysis];
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Accident
Investigation

Failure Traceability

Complaint
Analysis

Fault-tree Analysis

Brief explanation of each in the following slides

Accident Investigation

- Large-scale national disasters like oil tanker spillages and aircraft accidents are investigated using specifically trained staff.
- In many senses, the reason so much attention goes into examining these kinds of failures after the event is not only because of the damaging consequences of failure, but also because their infrequency makes it relatively hard to identify new sources of risks in advance of an event.

Failure Traceability

- Where some businesses (either by choice or because of a legal requirement) adopt traceability procedures to ensure that all their failures are traceable.
- Any failures can be traced back to the process which produced them, the components from which they were produced, or the suppliers who provided them.

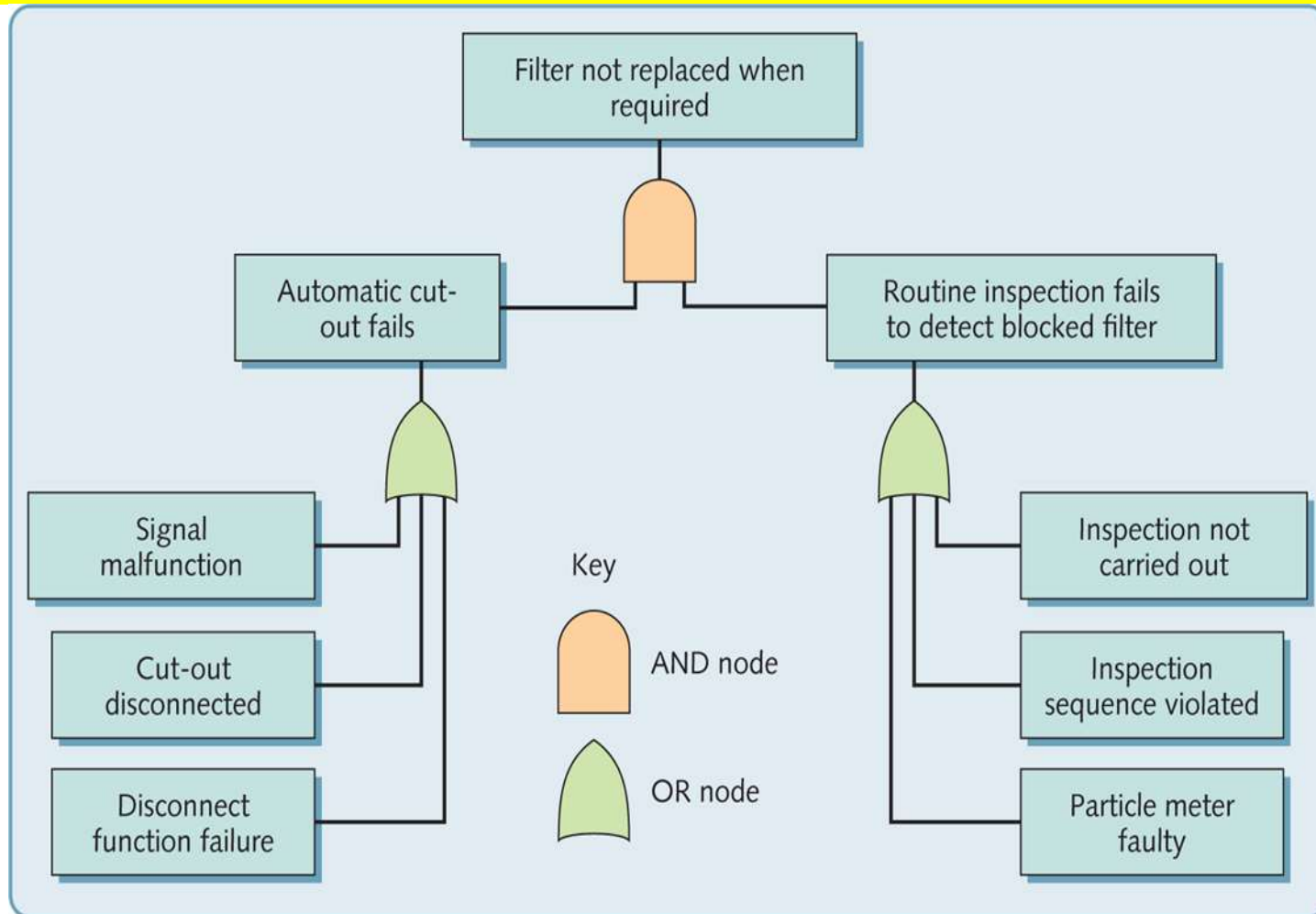
Complaint Analysis

- Where complaints are used as a valuable source for detecting the root causes of failures of customer service.
- The prime function of complaint analysis involves analysing the number and 'content' of complaints over time to understand better the nature of the failure, as the customer perceives it.
- Two key advantages of complaints are that they come unsolicited and they are often very timely pieces of information that can pinpoint problems quickly.

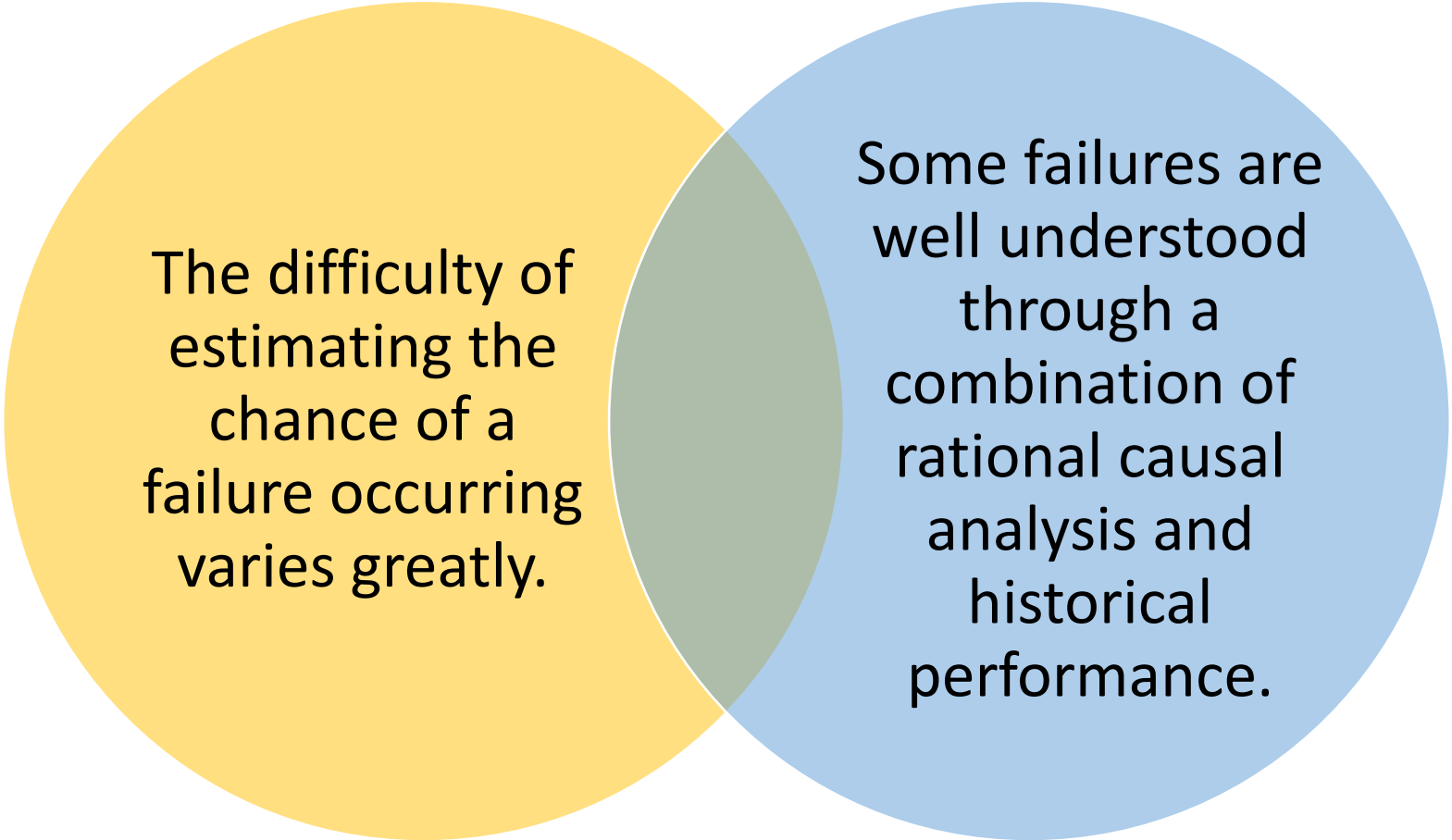
Fault-tree Analysis

- Where a logical procedure starts with a failure or a potential failure and works backwards to identify all the possible causes and therefore the origins of that failure.
- Fault-tree analysis is made up of branches connected by two types of nodes: AND nodes and OR nodes.
- The branches below an AND node all need to occur for the event above the node to occur.
- Only one of the branches below an OR node needs to occur for the event above the node to occur.

Fault-tree analysis for failure to replace filter when required



Likelihood of Failure



The difficulty of estimating the chance of a failure occurring varies greatly.

Some failures are well understood through a combination of rational causal analysis and historical performance.

Estimates of failure based on historical performance can be measured in three main ways:

- **Failure Rates** – how often a failure occurs;
- **Reliability** – the chances of a failure occurring;
- **Availability** – the amount of available useful operating time

Failure rate (FR) is calculated as the number of failures over a period of time

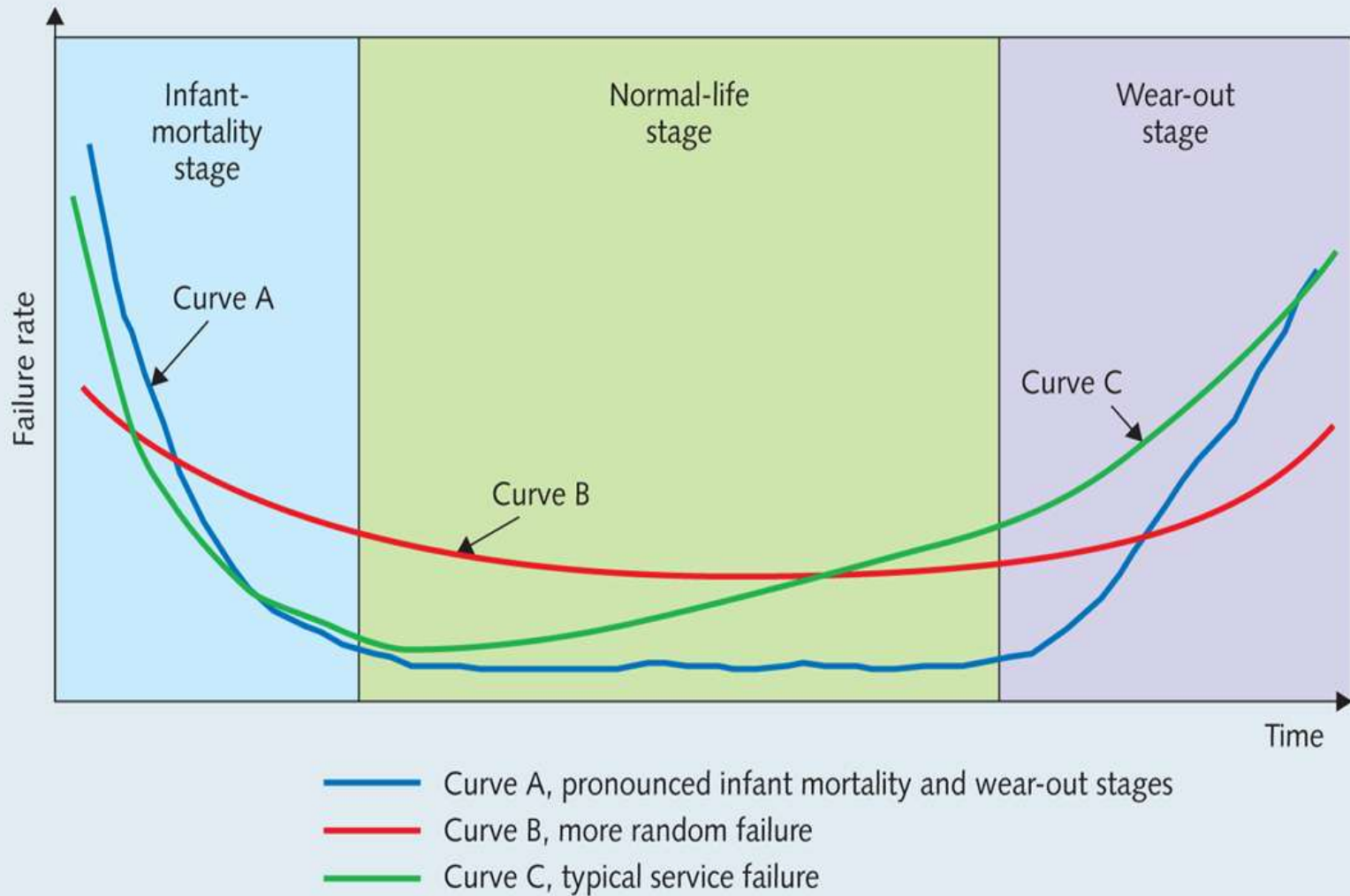
$$FR = \frac{\text{Number of failures}}{\text{Total number of products tested}} \times 100$$

$$FR = \frac{\text{Number of failures}}{\text{Operating time}}$$

Bath-tub curves

- The curve that describes failure probability of this type is called the bath-tub curve.
- It comprises three distinct stages:
 - the '**infant-mortality**' or '**early-life**' stage where early failures occur caused by defective parts or improper use;
 - the '**normal-life**' stage when the failure rate is usually low and reasonably constant, and caused by normal random factors;
 - and the '**wear-out**' stage when the failure rate increases as the part approaches the end of its working life and failure is caused by the ageing and deterioration of parts.

Bath-tub curves



Reliability failures

- Reliability measures the ability to perform as expected over time.
- Usually the importance of any particular failure is determined partly by how interdependent the other parts of the system are.
- With interdependence, a failure in one component will cause the whole system to fail.

Availability failures

- Availability is the degree to which the operation is ready to work.
- An operation is not available if it has either failed or is being repaired following failure.
- There are several different ways of measuring it depending on how many of the reasons for not operating are included.
- Lack of availability because of planned maintenance or changeovers could be included, for example.
- When 'availability' is being used to indicate the operating time excluding the consequence of failure

Mini Case Study



Activity

15 minutes

OPERATIONS IN PRACTICE

The rise of the MicroMort⁵

Calculating risk is far from easy. However, two authorities on risk, Michael Blastand and David Spiegelhalter, try to untangle the nature of risk and probability through the concept of the 'MicroMort'. The MicroMort is defined as a one-in-a-million chance of death and its use allows for some interesting comparisons to be made. For example, the chance of dying on a return motorbike trip from Edinburgh to London in the UK is around 120 MicroMorts (that is, a 120-in-a-million chance). This is comparable with the risk faced by mothers giving birth in the UK or a soldier spending 2.5 days in the most dangerous period of the Afghanistan conflict. The concept also allows us to examine how risks have changed over time. For example, aircrews involved in bombing raids during the Second World War were 'exposed' to 25,000 MicroMorts – a depressing 2.5 per cent chance of death – per mission. Soldiers in Afghanistan in the recent conflict faced (just) 47 Micromorts – or 0.0047 per cent chance of death – per day on the ground. What is perhaps most interesting about the work is that it explores



Source: 123RF.com: Andrew Mayovsky

just how irrational humans are when attempting to calculate risk. Typically high-impact/low-probability risk is over-emphasized, while low-impact/high-probability risk is under-emphasized. For organizations, appreciating the innate irrationality and cognition biases of most of its managers and employees can be an important step towards developing better assessments of various potential sources of failure.

Read the mini case study “The rise of the MicroMort” Make your points and discuss them with your peers

Failure Mode and Effect Analysis [FMEA]

- Having identified potential sources of failure (either in advance of an event or through post-failure analysis) and having then examined the likelihood of these failures occurring through some combination of objective and subjective analysis, managers can move to assigning relative priorities to risk.
- The most well-known approach for doing this is failure mode and effect analysis (FMEA).
- Its objective is to identify the factors that are critical to various types of failure as a means of identifying failures before they happen.

FMEA

What is the likelihood that failure will occur?

What would the consequence of the failure be?

How likely is such a failure to be detected before it affects the customer?

HOW CAN FAILURES BE PREVENTED?

'Prevention' is better than 'cure', which is why failure prevention is such an important responsibility for operations managers.

The obvious way to do this is to examine systematically any processes involved and 'design out' any failure points.

Many of the approaches such as product and service innovation, process design and quality management have been covered earlier and in this session, we will look at three further approaches to reducing risk

Preventing Failure

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graph LR; A[Preventing Failure] --- B[Building redundancy into a process in case of failure]; A --- C['Fail-safing' some of the activities in the process]; A --- D[Maintaining the physical facilities in the process];
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Building redundancy into a process in case of failure

‘Fail-safing’ some of the activities in the process

Maintaining the physical facilities in the process



Do you remember?



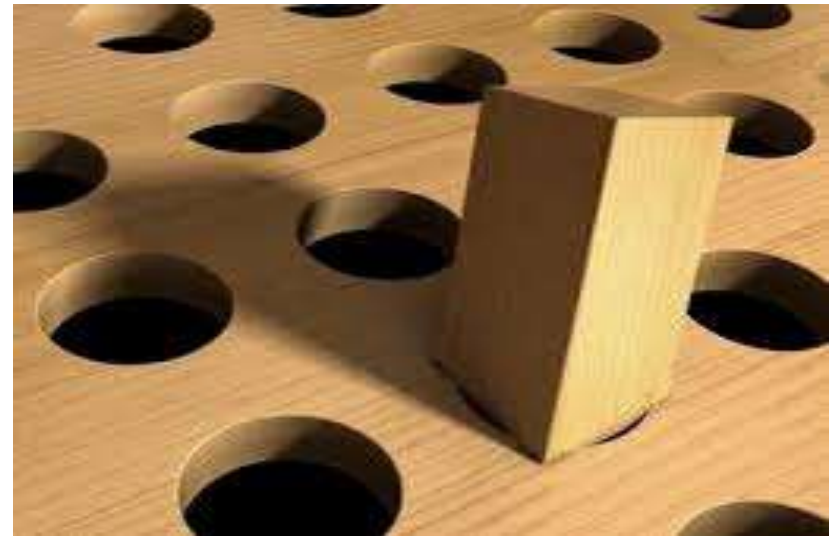
Activity

10 minutes

- Do you remember Poka-yoke?
- Give yourself a few minutes to remember
- And share your learning on AULA
- Next, check the previous sessions of the week
- Finally, share your reflection again on AULA



- The concept of **fail-safeing** has emerged since the introduction of Japanese methods of operations improvement.
- Called poka-yoke in Japan (from yokeru (to prevent) and poka (inadvertent errors)), the **idea is based on the principle** that human mistakes are to some extent inevitable.
- What is important is to **prevent them becoming defects**.
- Poka-yokes are simple (preferably inexpensive) devices or systems that are incorporated into a process to prevent inadvertent mistakes by those providing a service as well as customers receiving a service.



Examples of poka-yokes include:

Trays used in hospitals with indentations shaped to each item needed for a surgical procedure – any item not back in place at the end of the procedure might have been left in the patient;

Checklists which have to be filled in, either in preparation for, or on completion of, an activity, such as a maintenance checklist for an aircraft during turnaround;

Gauges placed on machines through which a part has to pass in order to be loaded onto, or taken off, the machine – an incorrect size or orientation stops the process;

The locks on aircraft lavatory doors, which must be turned to switch the light on;

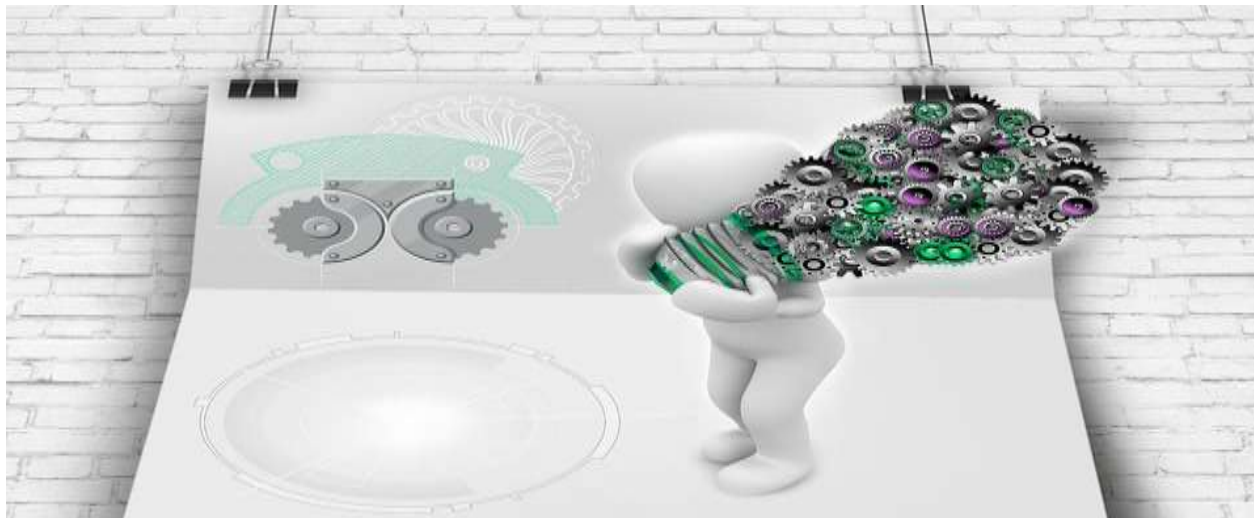
Beepers on ATMs to ensure that customers remove their cards or in cars to remind drivers to take their keys with them;

Limit switches on machines which allow the machine to operate only if the part is positioned correctly;

Height bars on amusement rides to ensure that customers do not exceed size limitations.

Maintenance

- **Maintenance** is how organizations try **to avoid failure** by taking care of their physical facilities.
- It is an important part of most operations' activities, particularly in operations dominated by their physical facilities such as power stations, hotels, airlines and petrochemical refineries.
- The **benefits of effective** maintenance include enhanced safety, increased reliability, higher quality, lower operating costs, a longer life span for process technology, and higher 'end value' (because well-maintained facilities are generally easier to dispose of into the second-hand market).



Total productive maintenance [TPM]

TPM is 'the productive maintenance carried out by all employees through small-group activities', where productive maintenance is, 'maintenance management which recognizes the importance of reliability, maintenance and economic efficiency in plant design'

Five goals of TPM:

1. Improve equipment effectiveness by examining all the losses which occur.
2. Achieve autonomous maintenance by allowing staff to take responsibility for some of the maintenance tasks and for the improvement of maintenance performance.
3. Plan maintenance with a fully worked-out approach to all maintenance activities.
4. Train all staff in relevant maintenance skills so that both maintenance and operating staff have all the skills to carry out their roles.
5. Achieve early equipment management by 'maintenance prevention' (MP), which involves considering failure causes and the maintainability of equipment during its design, manufacture, installation and commissioning.

MITIGATE THE EFFECTS OF FAILURE

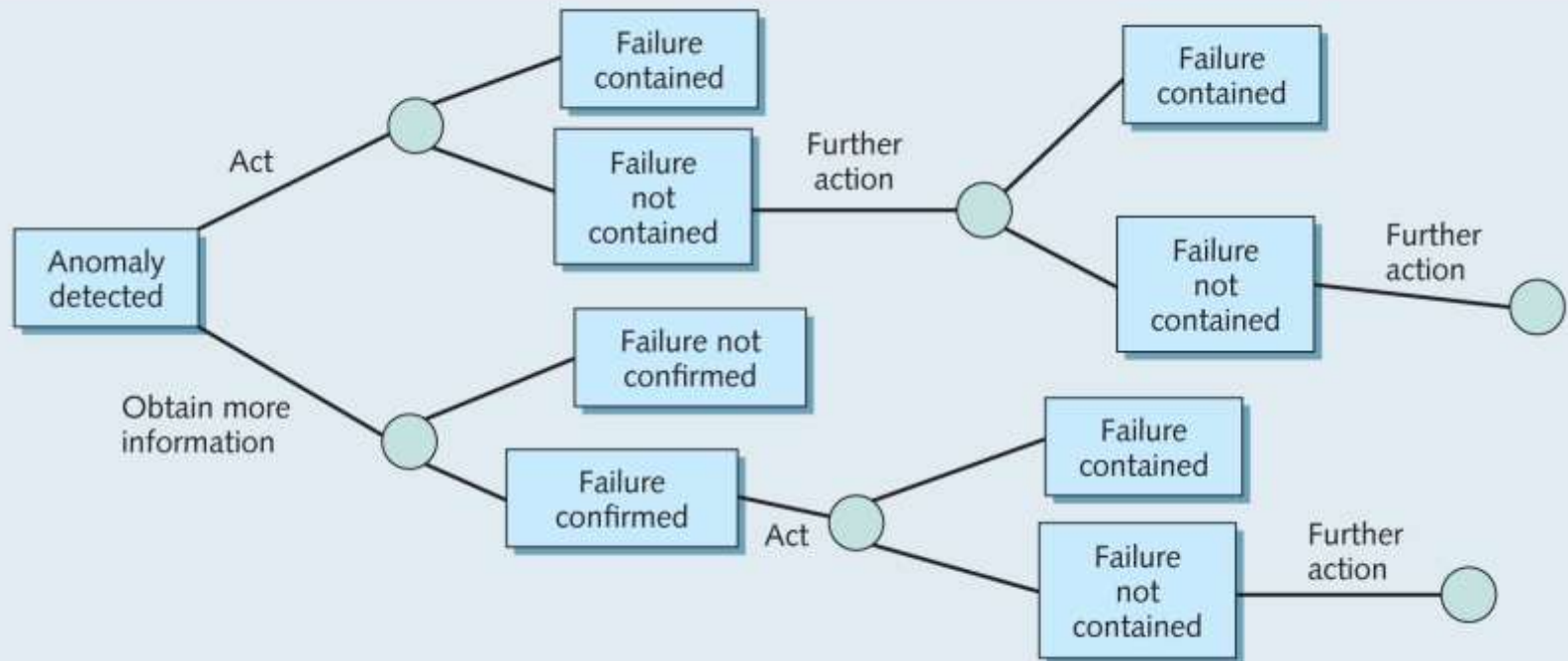
Failure (or risk) mitigation means isolating a failure from its negative consequences.

- It is an admission that not all failures can be avoided.
- However, in some areas of operations management relying on mitigation, rather than prevention, is unfashionable.
- For example, 'inspection' practices in quality management were based on the assumption that failures were inevitable and they needed to be detected before they could cause harm.

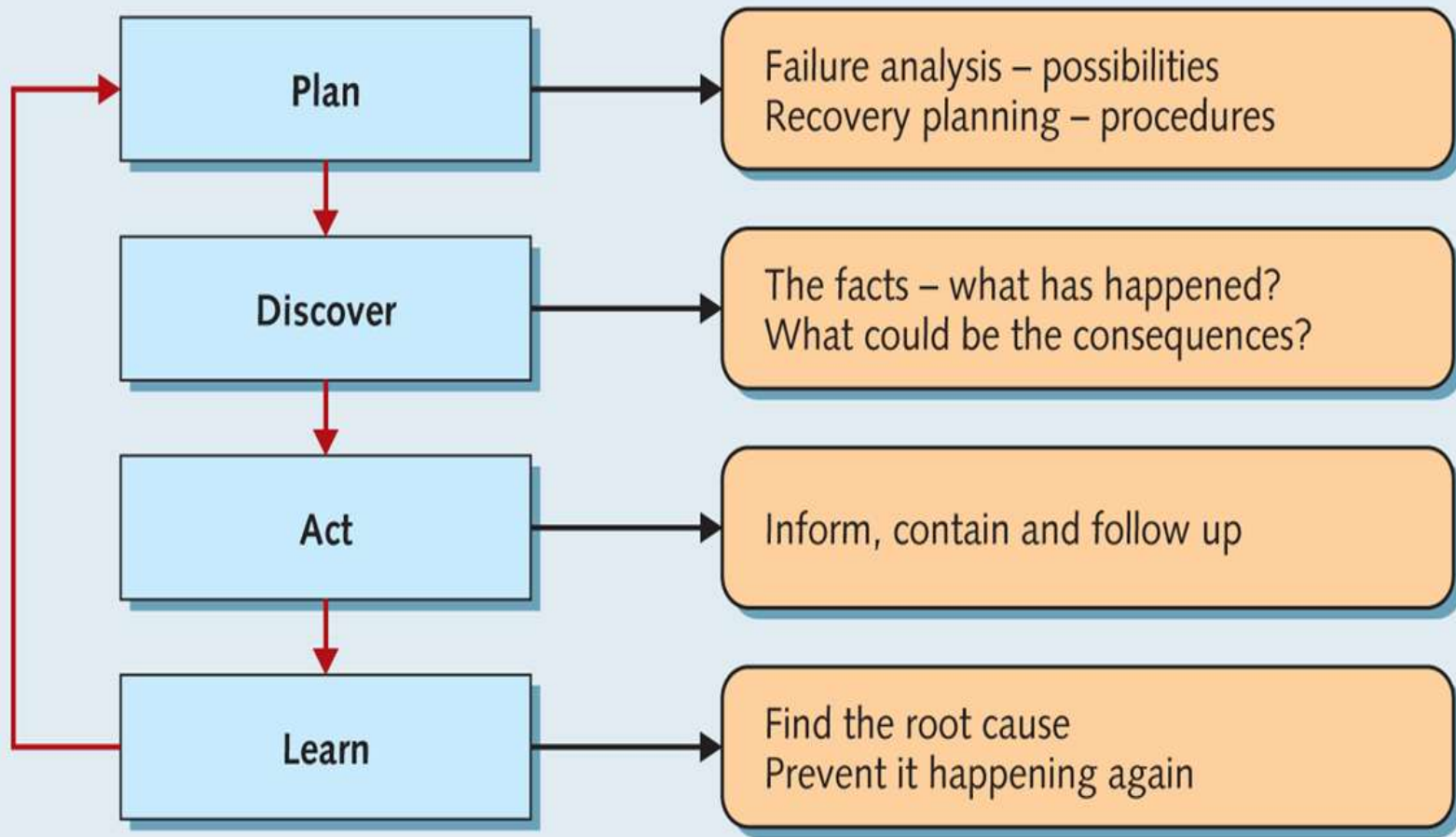
Failure Mitigation Actions

- Mitigation planning
- Economic mitigation
- Containment (spatial)
- Containment (temporal)
- Loss reduction
- Substitution

A decision tree for mitigation when failure is not immediately obvious



Recovery sequence for minimising the impact from failure



Mini Case Study



Activity

15 minutes

OPERATIONS IN PRACTICE

Tiny and not so tiny recalls for General Motors⁸

According to the data analytics company Teradata Corp., car recalls cost the automotive industry somewhere between \$45 and \$50 billion per annum. Which is why automotive giants including General Motors, Nissan and Toyota have looked to build sophisticated data-mining operations so they can move towards what industry insiders call 'tiny recalls'. When faced with an emerging potential mechanical problem, car manufacturers have traditionally had two choices. The first option is to issue a general recall for the model in question. But there is a major financial and reputational cost to this, as well as significant inconvenience for customers, many of who will not actually have a fault in their car. The second option is to ignore the fault and in doing so accept the risk that a very small number of customers may have problems (even serious accidents). While this may be justified in short-term financial (if not ethical) terms, the longer term consequences could be disastrous, especially if the media expose the fact. Now, there is a third option available. The rise of bar coding, radio frequency identification (RFID) and more advanced information systems has allowed automakers to track components in far greater detail. The result is that faults can often be isolated not only to a model range, but to a specific set of components, made in particular periods of time, even by a particular shift at the factory. This then enables a far



Source: Corbis, Michael Spooner/Bettar

more focused recall of only the cars that are most likely to have a defect. For example, when a braking fault was identified on the Chevrolet Volt, General Motors (GM) was able to identify all 'at-risk' cars through its advanced parts tracking, and issue a 'tiny recall' (in this case, just four cars in the whole of the USA). However, prevention is better than cure. Many auto manufacturers, while seeking to target recalls, still face large recalls brought about by quality problems. For example, during 2014, GM announced the recall of a staggering 29 million cars and vans in North America, a number greater than its total US sales between 2005 and 2013. One of the largest recalls was for ignition switch defects, which included models as old as 1997.

Read the mini case study
"Tiny and not so tiny
recalls for General
Motors"

Make your points and
discuss them with your
peers

Independent study



Read the article: Managing Risk Recovery Project Management

Assignment support



Traffic Light *it*

Where is your learning at?



Activity

5 minutes



Red = I have misunderstood some of the topics. (write down 2 questions that you need help with)

Amber = I have understood most of the topics so far (write down 1 question and 1 key theme you have understood)

Green = I have understood the topics so far (write down 2 key themes from the lesson to show your understanding)

Time for Reflection...



Activity

10 minutes

Reflect on what you have learned today and
share on AULA/Community Space:

What happened?

What were you thinking?

What were you feeling?

What was good / not good about today's
lecture?

Will it be useful?

A] Take down at least two points that you learned new

B] How you are going to transfer this learning in your CW
and/or workplace



*Thank
You*

Essential Reading

- Slack, N., Brandon-Jones, A. and Johnston, R. (2016) Operations Management. 8th edn. Harlow: Pearson
- Torrington, D., Hall, L., Atkinson, C. and Taylor, S. (2017) Human Resource Management. 10th edn. Harlow: Pearson

Recommended Reading

- Slack, N., Brandon-Jones, A., Johnston, R. and Betts, A. (2015) Operations and Process Management: Principles and Practice for Strategic Impact. 4th edn. Harlow: Pearson
- Bratton, J. and Gold, J. (2012) Human Resource Management: Theory and Practice. 5th edn. Basingstoke: Palgrave MacMillan
- Heizer, J., Render, B. and Munson, C. (2017) Principles of Operations Management: Sustainability and Supply Chain Management. 10th edn. Harlow: Pearson
- CMI Website : Management Direct

Session Reference List

- Greasley, A. (2013) Operations Management. 3rd edn. John Wiley & Sons
- Heizer, J., Render, Barry, author, & Munson, Chuck, author. (2017). Principles of operations management sustainability and supply chain management (Tenth edition.; Global ed.).
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- Slack, N., Brandon-Jones, Alistair, author, & Johnston, Robert, author. (2013). *Operations management* (Seventh ed.). Harlow, England: Pearson
- Slack, N. (2015) Operations and process management: principles and practice for strategic impact. Fourth edition. Harlow, England: Pearson Harlow, England: Pearson
- Slack, N. and Brandon-Jones, A. Johnston R. (2016) Operations Management. Eighth edition,
- Slack, N. and Brandon-Jones, A. (2018) Operations and process management: principles and practice for strategic impact. Fifth edition. Harlow, England: Pearson